

Orthogonal (Legendre) Polynomials Under (Laplace) Convolution

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For real w , define a convolution of the form $f * g = \int_0^w f(x) g(w-x) dx$; then there exist polynomials $Q_n(x, w)$ such that; (integers n, m)

$$Q_n(x, w) * Q_m(x, w) = 0 \quad \forall (n, m) \quad n \neq m$$

1 Heuristic Construction

By setting $Q_0(x, w) = 1$, let us generate successive polynomials one coefficient at a time by passing an arbitrary polynomial of degree n through each of the lower polynomials to ensure a convolution which equates to 0. Solving the n resulting polynomial equations for the $n-1$ coefficients (assume x^n has a coefficient of 1) then pins Q_n exactly.

This method produces;

$$\begin{aligned} Q_1(x, w) &= x - \frac{w}{2} \\ Q_2(x, w) &= x^2 - wx + \frac{w^2}{6} \\ Q_3(x, w) &= x^3 - \frac{3}{2}wx^2 + \frac{3}{5}w^2x - \frac{1}{20}w^3 \\ &\dots \end{aligned}$$

Next, by normalizing by the evident central binomial coefficient of order n we achieve:

n	$\binom{2n}{n} Q_n(x, w) = R_n(x, w)$
0	1
1	$2x - w$
2	$6x^2 - 6wx + w^2$
3	$20x^3 - 30wx^2 + 12w^2x - w^3$
4	$70x^4 - 140wx^3 + 90w^2x^2 - 20w^3x + w^4$
5	$252x^5 - 630wx^4 + 560w^2x^3 - 210w^3x^2 + 30w^4x - w^5$
6	$924x^6 - 2772wx^5 + 3150w^2x^4 - 1680w^3x^3 + 420w^4x^2 - 42w^5x + w^6$

Table 1: Scaled orthogonal polynomials $R_n(x, w)$.

Churning these integers through the OEIS, the sequence of *Delannoy numbers* appears. Using their explicit formula and the patterns seen in Table 1, we receive:

$$Q_n(x, w) = \frac{1}{\binom{2n}{n}} \sum_{k=0}^n x^{(n-k)} w^k \binom{n}{k} \binom{2n-k}{n} (-1)^k \quad (1)$$

2 Properties

A very important family of orthogonal polynomials are the *Legendre Polynomials*, which being defined as $P_n(x) = \sum_{k=0}^n \text{nCr}(n, k) \text{nCr}(n+k, k) \left(\frac{x-1}{2}\right)^k$, when viewed with powers reversed, produces a more recognizable form of;

$$R_n(x, w) = P_n\left(\frac{-2x}{w} + 1\right) (-w)^n$$

By the generating function for Legendre polynomials we receive the immediate

$$\sum_{n=0}^{\infty} R_n(x, w) t^n = \frac{1}{\sqrt{1 + 2(w - 2x)t + w^2 t^2}}$$

Also as a result of the Legendre polynomials, orthogonality under convolution boils down to a trivial substitution using $P_n(-x) = (-1)^n P_n(x)$;

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0_{n \neq m}$$

$$\int_0^w P_n\left(\frac{2}{w}x - 1\right) P_m\left(\frac{2}{w}x - 1\right) dx = 0_{n \neq m}$$

$$\int_0^w P_n\left(\frac{2}{w}x - 1\right) P_m\left(\frac{2}{w}(w - x) - 1\right) dx (-1)^m = 0_{n \neq m}$$

A simple fact significantly less dazzling than ‘mysterious’ progressively generated polynomials. All this concludes the proof that;

$$\int_0^w R_n(x, w) R_m(w - x, w) dx = 0_{n \neq m}$$